Map Scales and Directions

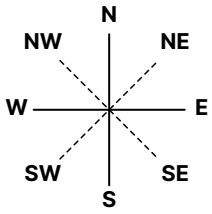
Name _____ **Date** _____ No calculator Version 1

KEY WORDS compass point · clockwise · anticlockwise · turn · scale · grid reference · represents

A Compass points and turns ●○○ — ANSWER ON THE SHEET

> **Remember:** There are 45° between one compass point and the next. A quarter turn is 90°, a half turn is 180° and a full turn is 360°.

1. Which way do you face after each turn?



a) You face north and turn 90° clockwise. _____

b) You face east and turn 180° clockwise. _____

c) You face south and turn 90° anticlockwise. _____

d) You face north and turn 45° clockwise. _____

e) You face west and turn 135° clockwise. _____

f) You face south-east and turn 90° anticlockwise. _____

2. What angle do you turn through, clockwise?

a) From north to east _____ **b)** From north to south-east _____

c) From east to west _____ **d)** From south to north-west _____

e) From north-east to south _____ **f)** From west to south _____

3. Write the direction that is opposite to each of these.

a) North _____ **b)** South-east _____ **c)** West _____ **d)** North-east _____

B Map scales ●●○ — WORK IN YOUR BOOK

> **Remember:** A scale tells you what 1 cm on the map stands for. If 1 cm represents 5 km, then 4 cm represents $4 \times 5 = 20$ km.

4. Find the real distance.

a) 1 cm represents 5 km. A distance on the map is 4 cm.

b) 1 cm represents 2 km. A distance on the map is 7.5 cm.

c) 1 cm represents 50 m. A distance on the map is 6 cm.

d) 1 cm represents 20 km. A distance on the map is 3.5 cm.

5. Find the distance on the map.

a) 1 cm represents 5 km. A real distance is 35 km.

b) 1 cm represents 2 km. A real distance is 13 km.

c) 1 cm represents 50 m. A real distance is 450 m.

d) 1 cm represents 20 km. A real distance is 150 km.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C Scales written as ratios ●●● — WORK IN YOUR BOOK

⚠ **Watch out:** A scale of 1 : 50,000 means 1 cm on the map is 50,000 cm in real life. Change the answer to metres (÷ 100) or kilometres (÷ 100,000).

6. Answer each question.

a) The scale is 1 : 1,000. A distance on the map is 6 cm. Find the real distance in metres.

b) The scale is 1 : 50,000. A distance on the map is 3 cm. Find the real distance in kilometres.

c) The scale is 1 : 200. A distance on the map is 4.5 cm. Find the real distance in metres.

d) The scale is 1 : 100,000. A distance on the map is 7.5 cm. Find the real distance in kilometres.

D Grid references ●●○ — ANSWER ON THE SHEET

Remember: A grid reference names a square: the letter first (across), then the number (up). North is at the top of the map.

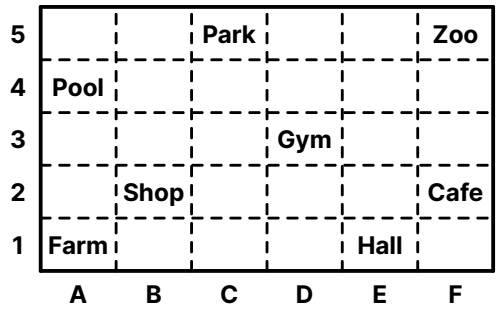
7. Use the map. Write the grid reference of each place.

a) The gym _____

b) The pool _____

c) The zoo _____

d) The farm _____



8. Which place is in each square?

a) Square C5 _____ **b)** Square B2 _____

c) Square E1 _____ **d)** Square F2 _____

9. Follow each set of directions on the map.

a) Start at the shop. Go 2 squares east and then 1 square north. Where are you? _____

b) Start at the zoo. Go 3 squares west. Where are you? _____

c) Start at the farm. Go 3 squares north. Where are you? _____

d) Start at the cafe. Go 1 square west and then 1 square south. Where are you? _____

★ Extension — FOR EARLY FINISHERS

10. The scale is 1 : 25,000. A distance on the map is 8 cm. Find the real distance in kilometres.

11. A walking track is 14 km long. How long is it on a map with a scale of 1 : 50,000?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Map Scales and Directions

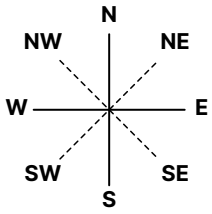
Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS compass point · clockwise · anticlockwise · turn · scale · grid reference · represents

A Compass points and turns ●○○ — ANSWER ON THE SHEET

> **Remember:** There are 45° between one compass point and the next. A quarter turn is 90°, a half turn is 180° and a full turn is 360°.

1. Which way do you face after each turn?



a) You face north and turn 90° anticlockwise. _____

b) You face west and turn 180° clockwise. _____

c) You face east and turn 90° clockwise. _____

d) You face north and turn 45° anticlockwise. _____

e) You face south and turn 135° clockwise. _____

f) You face north-east and turn 90° clockwise. _____

2. What angle do you turn through, clockwise?

a) From north to south _____ **b)** From north to north-east _____

c) From east to south _____ **d)** From west to north-east _____

e) From south-east to west _____ **f)** From south to east _____

3. Write the direction that is opposite to each of these.

a) South _____ **b)** South-west _____ **c)** East _____ **d)** North-west _____

B Map scales ●●○ — WORK IN YOUR BOOK

> **Remember:** A scale tells you what 1 cm on the map stands for. If 1 cm represents 5 km, then 4 cm represents $4 \times 5 = 20$ km.

4. Find the real distance.

a) 1 cm represents 4 km. A distance on the map is 6 cm.

b) 1 cm represents 2 km. A distance on the map is 8.5 cm.

c) 1 cm represents 50 m. A distance on the map is 8 cm.

d) 1 cm represents 20 km. A distance on the map is 4.5 cm.

5. Find the distance on the map.

a) 1 cm represents 4 km. A real distance is 36 km.

b) 1 cm represents 2 km. A real distance is 15 km.

c) 1 cm represents 50 m. A real distance is 350 m.

d) 1 cm represents 20 km. A real distance is 130 km.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C Scales written as ratios ●●● — WORK IN YOUR BOOK

⚠ **Watch out:** A scale of 1 : 50,000 means 1 cm on the map is 50,000 cm in real life. Change the answer to metres (÷ 100) or kilometres (÷ 100,000).

6. Answer each question.

a) The scale is 1 : 1,000. A distance on the map is 8 cm. Find the real distance in metres.

b) The scale is 1 : 50,000. A distance on the map is 5 cm. Find the real distance in kilometres.

c) The scale is 1 : 200. A distance on the map is 3.5 cm. Find the real distance in metres.

d) The scale is 1 : 100,000. A distance on the map is 4.5 cm. Find the real distance in kilometres.

D Grid references ●●○ — ANSWER ON THE SHEET

Remember: A grid reference names a square: the letter first (across), then the number (up). North is at the top of the map.

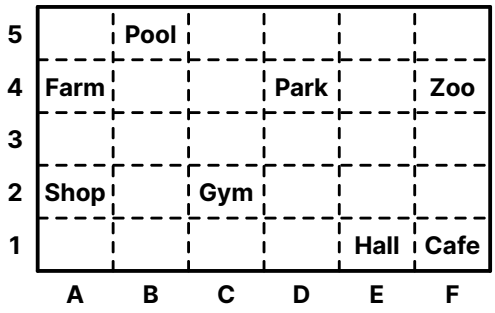
7. Use the map. Write the grid reference of each place.

a) The gym _____

b) The pool _____

c) The zoo _____

d) The farm _____



8. Which place is in each square?

a) Square D4 _____ **b)** Square A2 _____

c) Square E1 _____ **d)** Square F1 _____

9. Follow each set of directions on the map.

a) Start at the shop. Go 2 squares east. Where are you? _____

b) Start at the zoo. Go 2 squares west. Where are you? _____

c) Start at the hall. Go 1 square east. Where are you? _____

d) Start at the farm. Go 1 square east and then 1 square north. Where are you? _____

★ Extension — FOR EARLY FINISHERS

10. The scale is 1 : 25,000. A distance on the map is 12 cm. Find the real distance in kilometres.

11. A walking track is 9 km long. How long is it on a map with a scale of 1 : 50,000?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.